WEEK 3: SUPERMARKET MANAGEMENT SYSTEM

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Scenario

The Supermarket Management System needs to maintain supplier and inventory information. The system stores supplier details and tracks the stock quantity, stock status, and latest inventory update date for products available in the supermarket.

The system helps the supermarket monitor product availability and supplier information efficiently.

Creating Supplier and Inventory Tables

Two tables are created for this project:

SUPPLIER Table

The **SUPPLIER** table stores information about suppliers who provide products to the supermarket.

Attribute	Description
SUPPLIER_ID	Unique ID of the supplier
SUPPLIER_NAME	Name of the supplier
PHONE	Supplier contact number
EMAIL	Supplier email address

```
SQL> CREATE TABLE SUPPLIER (
  2      SUPPLIER_ID VARCHAR2(10) PRIMARY KEY,
  3      SUPPLIER_NAME VARCHAR2(100) NOT NULL,
  4      PHONE VARCHAR2(15) NOT NULL,
  5      EMAIL VARCHAR2(100) NOT NULL UNIQUE
  6  );

Table created.

SQL> DESC SUPPLIER;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------
 SUPPLIER_ID                               NOT NULL VARCHAR2(10)
 SUPPLIER_NAME                             NOT NULL VARCHAR2(100)
 PHONE                                     NOT NULL VARCHAR2(15)
 EMAIL                                     NOT NULL VARCHAR2(100)

SQL>
```

INVENTORY Table

The INVENTORY table stores information about the stock available for supermarket products.

Attribute	Description
INVENTORY_ID	Unique ID of the inventory record
PRODUCT_ID	Product associated with the inventory
SUPPLIER_ID	Supplier providing the product
STOCK_QUANTITY	Available quantity
STOCK_STATUS	Current stock status
LAST_UPDATED	Date of latest inventory update

```
SQL> CREATE TABLE INVENTORY (
  2      INVENTORY_ID VARCHAR2(10) PRIMARY KEY,
  3      PRODUCT_ID VARCHAR2(10) NOT NULL,
  4      SUPPLIER_ID VARCHAR2(10) NOT NULL,
  5      STOCK_QUANTITY NUMBER(10) NOT NULL,
  6      STOCK_STATUS VARCHAR2(20) NOT NULL,
  7      LAST_UPDATED DATE NOT NULL,
  8      CONSTRAINT FK_INVENTORY_PRODUCT
  9          FOREIGN KEY (PRODUCT_ID)
 10          REFERENCES PRODUCT(PRODUCT_ID),
 11      CONSTRAINT FK_INVENTORY_SUPPLIER
 12          FOREIGN KEY (SUPPLIER_ID)
 13          REFERENCES SUPPLIER(SUPPLIER_ID),
 14      CONSTRAINT CHK_INVENTORY_STOCK
 15          CHECK (STOCK_QUANTITY > 0)
 16  );

Table created.

SQL> DESC INVENTORY;
 Name                                      Null?    Type
 ----------------------------------------- -------- ------------
 INVENTORY_ID                              NOT NULL VARCHAR2(10)
 PRODUCT_ID                                NOT NULL VARCHAR2(10)
 SUPPLIER_ID                               NOT NULL VARCHAR2(10)
 STOCK_QUANTITY                            NOT NULL NUMBER(10)
 STOCK_STATUS                              NOT NULL VARCHAR2(20)
 LAST_UPDATED                              NOT NULL DATE

SQL>
```

Establish Relationship Using Primary Key and Foreign Key

Primary Key Relationship

The tables use primary keys to uniquely identify each record:

- SUPPLIER_ID → Primary Key in SUPPLIER
- INVENTORY_ID → Primary Key in INVENTORY
- PRODUCT_ID → Primary Key in your existing PRODUCT table from Week 2

Foreign Key Relationship

The INVENTORY table is connected to the existing PRODUCT and SUPPLIER tables using foreign keys.

PRODUCT

|

| PRODUCT_ID

|

↓

INVENTORY

↑

| SUPPLIER_ID

|

SUPPLIER

The relationships are:

PRODUCT.PRODUCT_ID

↓

INVENTORY.PRODUCT_ID

SUPPLIER.SUPPLIER_ID

↓

INVENTORY.SUPPLIER_ID

```
SQL> CREATE TABLE INVENTORY (
  2      INVENTORY_ID VARCHAR2(10) PRIMARY KEY,
  3      PRODUCT_ID VARCHAR2(10) NOT NULL,
  4      SUPPLIER_ID VARCHAR2(10) NOT NULL,
  5      STOCK_QUANTITY NUMBER(10) NOT NULL,
  6      STOCK_STATUS VARCHAR2(20) NOT NULL,
  7      LAST_UPDATED DATE NOT NULL,
  8      CONSTRAINT FK_INVENTORY_PRODUCT
  9          FOREIGN KEY (PRODUCT_ID)
 10          REFERENCES PRODUCT(PRODUCT_ID),
 11      CONSTRAINT FK_INVENTORY_SUPPLIER
 12          FOREIGN KEY (SUPPLIER_ID)
 13          REFERENCES SUPPLIER(SUPPLIER_ID),
 14      CONSTRAINT CHK_INVENTORY_STOCK
 15          CHECK (STOCK_QUANTITY > 0)
 16  );

Table created.

SQL> DESC INVENTORY;
 Name                                      Null?    Type
 ----------------------------------------- -------- ------------
 INVENTORY_ID                              NOT NULL VARCHAR2(10)
 PRODUCT_ID                                NOT NULL VARCHAR2(10)
 SUPPLIER_ID                               NOT NULL VARCHAR2(10)
 STOCK_QUANTITY                            NOT NULL NUMBER(10)
 STOCK_STATUS                              NOT NULL VARCHAR2(20)
 LAST_UPDATED                              NOT NULL DATE

SQL>
```

Constraints

UNIQUE Constraint for Email

A UNIQUE constraint is added to the EMAIL column in the SUPPLIER table to ensure that two suppliers cannot have the same email address.

NOT NULL Constraint for Email

The NOT NULL constraint ensures that every supplier record must contain an email address.

CHECK Constraint for Stock Quantity

A CHECK constraint is added to the STOCK_QUANTITY column to ensure that the stock quantity is greater than zero.

STOCK_QUANTITY > 0

This prevents invalid stock quantities from being entered.

```
SQL> SELECT CONSTRAINT_NAME,
  2         CONSTRAINT_TYPE,
  3         STATUS
  4  FROM USER_CONSTRAINTS
  5  WHERE TABLE_NAME IN ('SUPPLIER', 'INVENTORY');

CONSTRAINT_NAME                C STATUS
------------------------------ - --------
SYS_C007186                    C ENABLED
SYS_C007187                    C ENABLED
SYS_C007188                    C ENABLED
SYS_C007189                    C ENABLED
SYS_C007190                    C ENABLED
CHK_INVENTORY_STOCK            C ENABLED
SYS_C007192                    P ENABLED
FK_INVENTORY_PRODUCT           R ENABLED
FK_INVENTORY_SUPPLIER          R ENABLED
SYS_C007181                    C ENABLED
SYS_C007182                    C ENABLED

CONSTRAINT_NAME                C STATUS
------------------------------ - --------
SYS_C007183                    C ENABLED
SYS_C007184                    P ENABLED
SYS_C007185                    U ENABLED

14 rows selected.
```

Inserting Values into SUPPLIER Table

Sample supplier records are inserted into the SUPPLIER table.

The records contain:

- Supplier ID
- Supplier name
- Phone number
- Email address

After insertion, the table contents are displayed to verify the records.

```
SQL> INSERT INTO SUPPLIER VALUES
  2  ('S001', 'Fresh Foods Suppliers', '9876543210', 'freshfoods@gmail.com');

1 row created.

SQL>
SQL> INSERT INTO SUPPLIER VALUES
  2  ('S002', 'Daily Needs Distributors', '9876543211', 'dailyneeds@gmail.com');

1 row created.

SQL>
SQL> INSERT INTO SUPPLIER VALUES
  2  ('S003', 'Green Basket Suppliers', '9876543212', 'greenbasket@gmail.com');

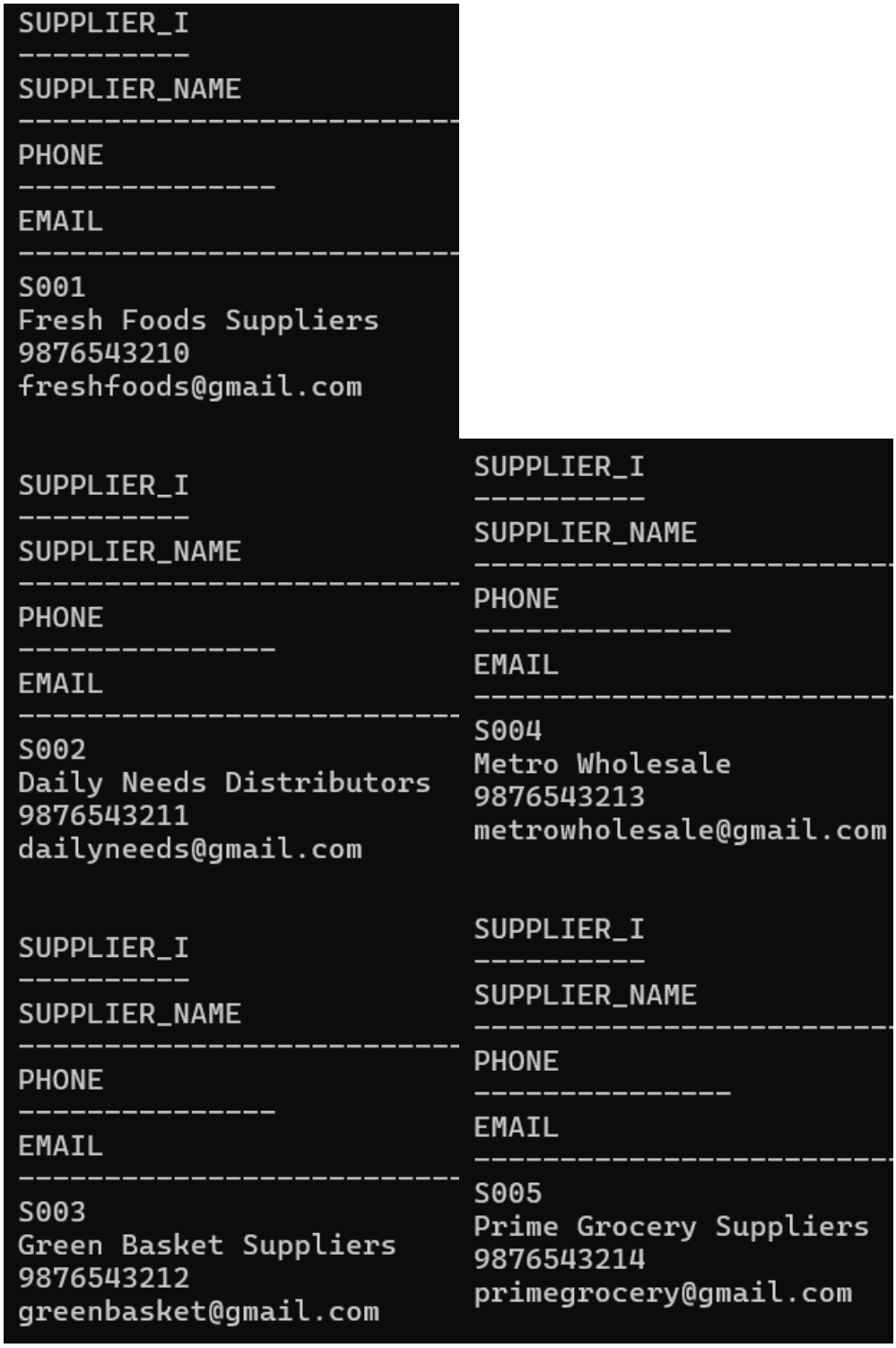
1 row created.

SQL>
SQL> INSERT INTO SUPPLIER VALUES
  2  ('S004', 'Metro Wholesale', '9876543213', 'metrowholesale@gmail.com');

1 row created.

SQL>
SQL> INSERT INTO SUPPLIER VALUES
  2  ('S005', 'Prime Grocery Suppliers', '9876543214', 'primegrocery@gmail.com');

1 row created.
```



Inserting Values into INVENTORY Table

Sample inventory records are inserted into the INVENTORY table. The records contain product, supplier, stock quantity, stock status, and the latest update date. The table contents are displayed to verify the inserted records.

```
SQL> INSERT INTO INVENTORY VALUES
  2  ('I001', 'P001', 'S001', 25, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I002', 'P002', 'S002', 30, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I003', 'P003', 'S003', 20, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I004', 'P004', 'S004', 15, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I005', 'P005', 'S005', 12, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I006', 'P006', 'S001', 50, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I007', 'P007', 'S002', 35, 'In Stock', SYSDATE);

1 row created.
```

```
SQL> INSERT INTO INVENTORY VALUES
  2  ('I008', 'P008', 'S003', 40, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I009', 'P009', 'S004', 25, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I010', 'P010', 'S005', 18, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I011', 'P011', 'S001', 30, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I012', 'P012', 'S002', 20, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I013', 'P013', 'S003', 16, 'In Stock', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I014', 'P014', 'S004', 14, 'In Stock', SYSDATE);
```

```
SQL>
SQL> INSERT INTO INVENTORY VALUES
  2  ('I015', 'P015', 'S005', 10, 'In Stock', SYSDATE);

1 row created.

SQL> COMMIT;

Commit complete.

SQL> SELECT * FROM INVENTORY;

INVENTORY_ PRODUCT_ID SUPPLIER_I STOCK_QUANTITY STOCK_STATUS         LAST_UPDA
---------- ---------- ---------- -------------- -------------------- ---------
I001       P001       S001                 25 In Stock             28-SEP-26
I002       P002       S002                 30 In Stock             28-SEP-26
I003       P003       S003                 20 In Stock             28-SEP-26
I004       P004       S004                 15 In Stock             28-SEP-26
I005       P005       S005                 12 In Stock             28-SEP-26
I006       P006       S001                 50 In Stock             28-SEP-26
I007       P007       S002                 35 In Stock             28-SEP-26
I008       P008       S003                 40 In Stock             28-SEP-26
I009       P009       S004                 25 In Stock             28-SEP-26
I010       P010       S005                 18 In Stock             28-SEP-26
I011       P011       S001                 30 In Stock             28-SEP-26

INVENTORY_ PRODUCT_ID SUPPLIER_I STOCK_QUANTITY STOCK_STATUS         LAST_UPDA
---------- ---------- ---------- -------------- -------------------- ---------
I012       P012       S002                 20 In Stock             28-SEP-26
I013       P013       S003                 16 In Stock             28-SEP-26
I014       P014       S004                 14 In Stock             28-SEP-26
I015       P015       S005                 10 In Stock             28-SEP-26

15 rows selected.
```

Supplier-Wise Product Report

This report displays the number of products associated with each supplier. The JOIN operation combines the SUPPLIER and INVENTORY tables, while COUNT and GROUP BY are used to calculate the number of products supplied by each supplier.

```
SQL> SELECT
  2       S.SUPPLIER_ID,
  3       S.SUPPLIER_NAME,
  4       COUNT(DISTINCT I.PRODUCT_ID) AS TOTAL_PRODUCTS
  5  FROM SUPPLIER S
  6  JOIN INVENTORY I
  7       ON S.SUPPLIER_ID = I.SUPPLIER_ID
  8  GROUP BY
  9       S.SUPPLIER_ID,
 10       S.SUPPLIER_NAME
 11  ORDER BY S.SUPPLIER_ID;

SUPPLIER_I
----------
SUPPLIER_NAME
--------------------------------------------------------------
TOTAL_PRODUCTS
--------------
S001
Fresh Foods Suppliers
              3

S002
Daily Needs Distributors
              3

SUPPLIER_I
----------
SUPPLIER_NAME
--------------------------------------------------------------
TOTAL_PRODUCTS
--------------

S003
Green Basket Suppliers
              3

S004
Metro Wholesale

SUPPLIER_I
----------
SUPPLIER_NAME
--------------------------------------------------------------
TOTAL_PRODUCTS
--------------
              3

S005
Prime Grocery Suppliers
              3
```

Stock Availability Report

This report displays the product name, available stock quantity, and current stock status.

Purpose

To check the current stock quantity and availability status of each supermarket product.

Information displayed

- Product ID
- Product Name
- Stock Quantity

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      I.STOCK_QUANTITY,
  5      I.STOCK_STATUS
  6  FROM PRODUCT P
  7  JOIN INVENTORY I
  8      ON P.PRODUCT_ID = I.PRODUCT_ID
  9  ORDER BY P.PRODUCT_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
STOCK_QUANTITY STOCK_STATUS
-------------- --------------------
P001
Rice 5kg
            25 In Stock

P002
Wheat Flour 1kg
            30 In Stock

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
STOCK_QUANTITY STOCK_STATUS
-------------- --------------------

P003
Sugar 1kg
            20 In Stock

P004
Orange Juice

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
STOCK_QUANTITY STOCK_STATUS
-------------- --------------------
            15 In Stock

P005
Green Tea
            12 In Stock

P006

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
STOCK_QUANTITY STOCK_STATUS
-------------- --------------------
Mineral Water
            50 In Stock

P007
Potato Chips
```



Inventory Analysis

This report analyzes the supermarket inventory and provides important stock information.

To Find:

➢ Total Available Products

The total number of products currently available in inventory.

```
SQL> SELECT COUNT(DISTINCT PRODUCT_ID) AS TOTAL_AVAILABLE_PRODUCTS
  2  FROM INVENTORY
  3  WHERE STOCK_QUANTITY > 0;

TOTAL_AVAILABLE_PRODUCTS
------------------------
                      15

SQL>
```

➢ Products Out of Stock

This query identifies products that are out of stock by checking for an inventory quantity equal to zero. The current inventory has no products with zero stock.

Products whose stock quantity is zero.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      I.STOCK_QUANTITY,
  5      I.STOCK_STATUS
  6  FROM PRODUCT P
  7  JOIN INVENTORY I
  8      ON P.PRODUCT_ID = I.PRODUCT_ID
  9  WHERE I.STOCK_QUANTITY = 0;

no rows selected

SQL>
```

➢ Highest Stock Product

This query identifies the product having the highest stock quantity in the supermarket inventory. The MAX function is used to find the maximum stock quantity, and the result is matched with the corresponding product.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      I.STOCK_QUANTITY
  5  FROM PRODUCT P
  6  JOIN INVENTORY I
  7      ON P.PRODUCT_ID = I.PRODUCT_ID
  8  WHERE I.STOCK_QUANTITY = (
  9      SELECT MAX(STOCK_QUANTITY)
 10      FROM INVENTORY
 11  );

PRODUCT_ID
----------
PRODUCT_NAME
----------------------------------------
STOCK_QUANTITY
--------------
P006
Mineral Water
           50


SQL>
```

➢ **Average Inventory Quantity**

The AVG function is used to calculate the average stock quantity available in the supermarket inventory. The ROUND function displays the result up to two decimal places.

```
SQL> SELECT
  2      ROUND(AVG(STOCK_QUANTITY), 2) AS AVERAGE_INVENTORY_QUANTITY
  3  FROM INVENTORY;

AVERAGE_INVENTORY_QUANTITY
--------------------------
                        24

SQL>
```

Conclusion

The Supermarket Management System efficiently manages supplier and inventory information using an Oracle database. The project demonstrates primary keys, foreign keys, UNIQUE, NOT NULL, and CHECK constraints, along with data insertion and SQL queries. The reports help analyze supplier-wise products, stock availability, highest stock, out-of-stock products, and average inventory quantity. The system helps maintain organized inventory records and supports effective supermarket stock management.